

Data Analytics — Complete Hinglish Summary + Mind Map Notes

1. Data

Definition

Data means raw facts and figures collected from different sources.

Examples:

- Marks = 90
 - Temperature = 42°C
 - Student names
 - Images and videos
-

2. Types of Data

(A) Structured Data

Fixed format me organized data.

Examples

- Excel sheets
- Database tables
- Bank records

Diagram

```
+-----+-----+-----+
| Roll No | Name   | Marks |
+-----+-----+-----+
| 101     | Adit   | 90    |
| 102     | Rahul  | 85    |
+-----+-----+-----+
```

Features

- Easy to search
 - Rows and columns format
 - Fast analysis
-

(B) Unstructured Data

Jiska koi fixed format nahi hota.

Examples

- Videos
- Images
- Emails
- Social media posts

Diagram

Image	Video	Audio	Text
*	*	*	*

Features

- Difficult to analyze
 - Huge amount of data
-

(C) Semi-Structured Data

Partially organized data.

Examples

- JSON
- XML
- HTML

Example JSON

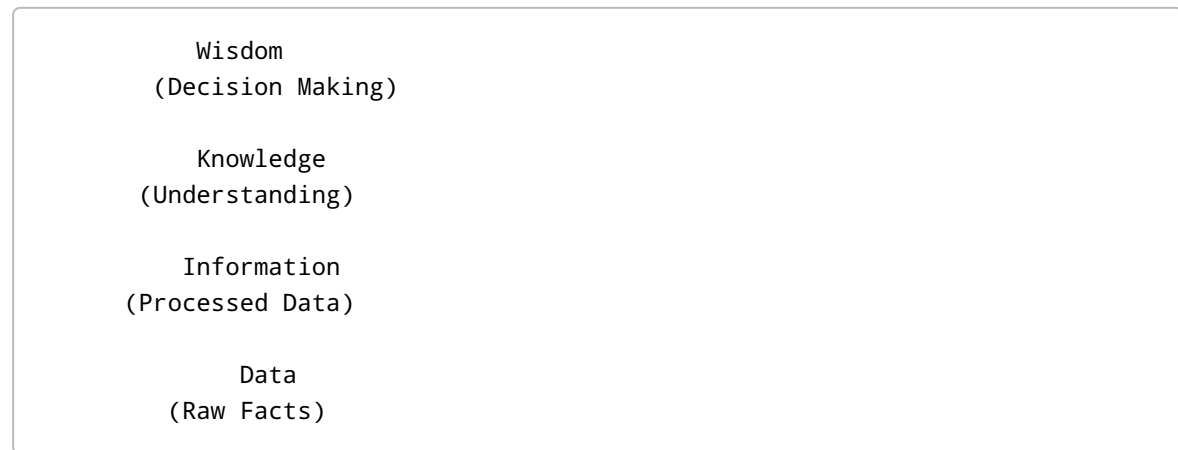
```
{  
  "name": "Adit",  
  "marks": 90  
}
```

Features

- Flexible structure
 - Easier than unstructured data
-

3. DIKW Model

Data → Information → Knowledge → Wisdom



Example

Data

85, 90, 78

Information

Average marks = 84.3

Knowledge

Student performance is good.

Wisdom

Student should get scholarship.

4. Characteristics of Data

Characteristic	Meaning
Accuracy	Correct data

Characteristic	Meaning
Completeness	No missing values
Consistency	Same format everywhere
Timeliness	Updated data
Relevance	Useful data
Reliability	Trusted source
Validity	Follows rules
Uniqueness	No duplicate records
Accessibility	Easy access
Security	Protected data

Important Characteristics

Accuracy

Correct and error-free data.

Example:

Marks = 90

Completeness

No important value should be missing.

Example:

Marks = NULL

Consistency

Same format everywhere.

Example:

14/05/2026
May 14, 2026

Timeliness

Latest and updated data.

Validity

Data should follow rules.

Example:

Age = -5

Invalid value.

5. Need of Data Analytics

Why Data Analytics is Needed?

1. Better Decision Making

Companies better decisions leti hain.

2. Customer Understanding

Customer behavior analyze hota hai.

3. Business Growth

Future trends predict hote hain.

4. Improving Efficiency

Time aur cost reduce hoti hai.

5. Risk Management

Fraud aur risks identify hote hain.

6. Future Prediction

Forecasting possible hoti hai.

6. Data Analytics Tools

Tool	Main Use
Excel	Basic analysis
Python	ML and analytics
R	Statistical analysis
SQL	Database queries
Tableau	Visualization
Power BI	Business intelligence
Hadoop	Big data
SAS	Enterprise analytics

Example Tools

Excel

- Charts
- Pivot tables
- Calculations

Python

Libraries:

- Pandas
- NumPy
- Matplotlib

SQL Example

```
SELECT * FROM students;
```

7. Applications of Data Analytics

Field	Application
Healthcare	Disease prediction
Banking	Fraud detection
E-commerce	Product recommendation
Education	Student analysis
Social Media	Personalized content
Transportation	Route optimization
Sports	Performance analysis
Weather	Forecasting

Real-Life Examples

Banking

Fraud transactions detect karna.

E-commerce

Products recommend karna.

Education

Weak students identify karna.

8. Data Analytics Lifecycle (DALC)

Phases of DALC

```
Discovery
  ↓
Data Preparation
  ↓
Model Planning
  ↓
Model Building
  ↓
Communicate Results
```

↓
Operationalize

1. Discovery

Problem identify karna.

Example: Sales decrease hone ka reason find karna.

2. Data Preparation

Data cleaning and organization.

Example: Missing values remove karna.

3. Model Planning

Techniques and algorithms choose karna.

Example: Regression model select karna.

4. Model Building

Model train aur test karna.

Example: Python me ML model banana.

5. Communicate Results

Reports aur charts create karna.

6. Operationalize

Model deployment and monitoring.

Example: Fraud detection system deploy karna.

9. Key Roles for Successful Analytic Projects

Role	Work
Business Analyst	Business requirements
Data Analyst	Data analysis and reports
Data Scientist	Machine learning models
Data Engineer	Data pipelines and storage
Project Manager	Team and project management
DBA	Database management

Important Roles

Business Analyst

Business goals identify karta hai.

Data Analyst

Reports and visualization banata hai.

Data Scientist

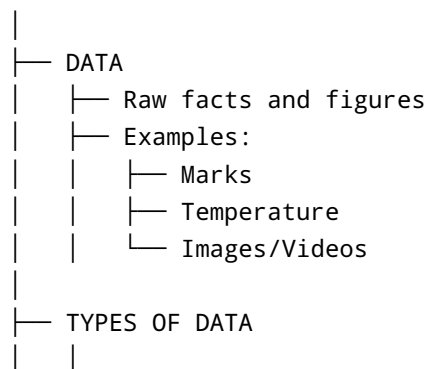
Prediction aur ML models banata hai.

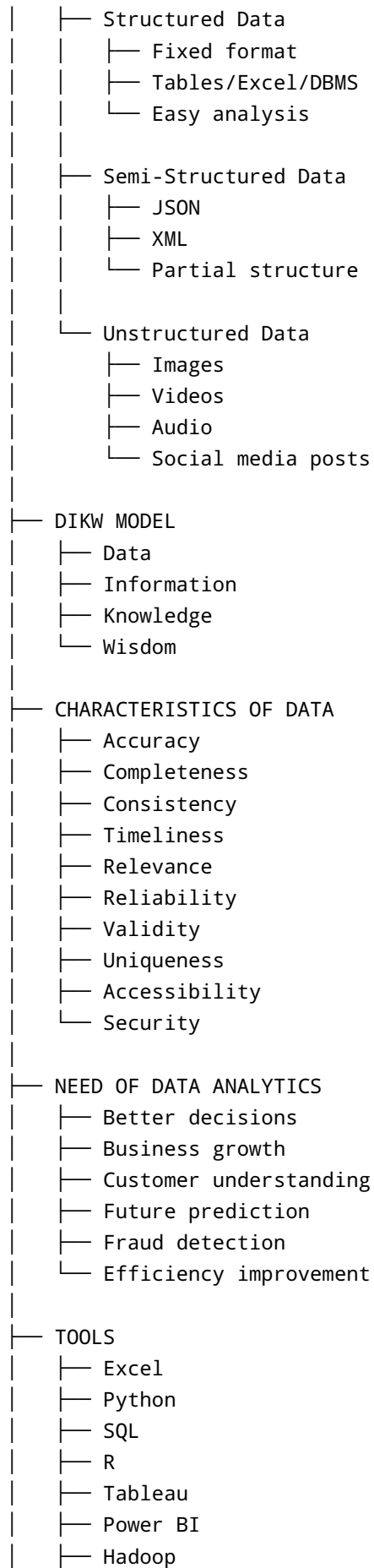
Data Engineer

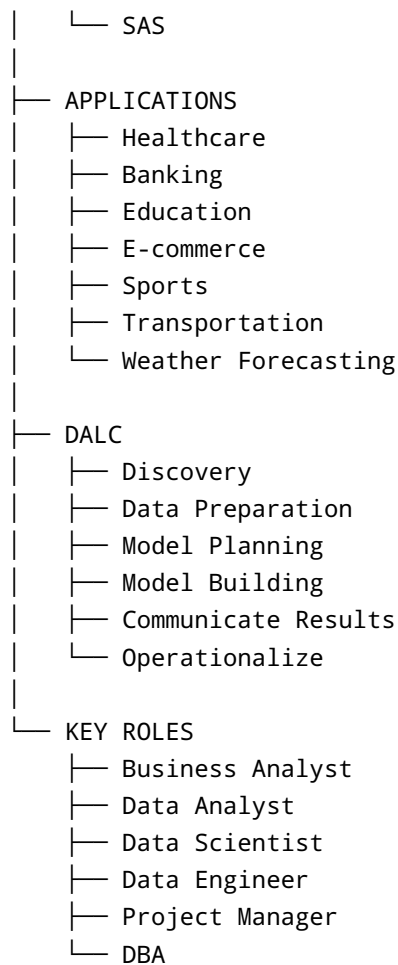
Data storage and processing systems manage karta hai.

FULL MIND MAP REVISION

DATA ANALYTICS







Quick Revision Sheet

Types of Data

Structured → Tables
Semi-Structured → JSON/XML
Unstructured → Images/Videos

DIKW

Data → Information → Knowledge → Wisdom

Important Characteristics

Accuracy
Completeness
Consistency
Timeliness
Validity

Important Tools

Excel
Python
SQL
Tableau
Power BI

Important Applications

Healthcare
Banking
E-commerce
Education
Transportation

DALC Steps

Discovery
Data Preparation
Model Planning
Model Building
Communicate Results
Operationalize

One-Line Definitions

Data

Raw facts and figures.

Information

Processed meaningful data.

Knowledge

Understanding from information.

Wisdom

Correct decision using knowledge.

Data Analytics

Process of analyzing data to get useful insights.

DALC

Step-by-step process of analytics project.